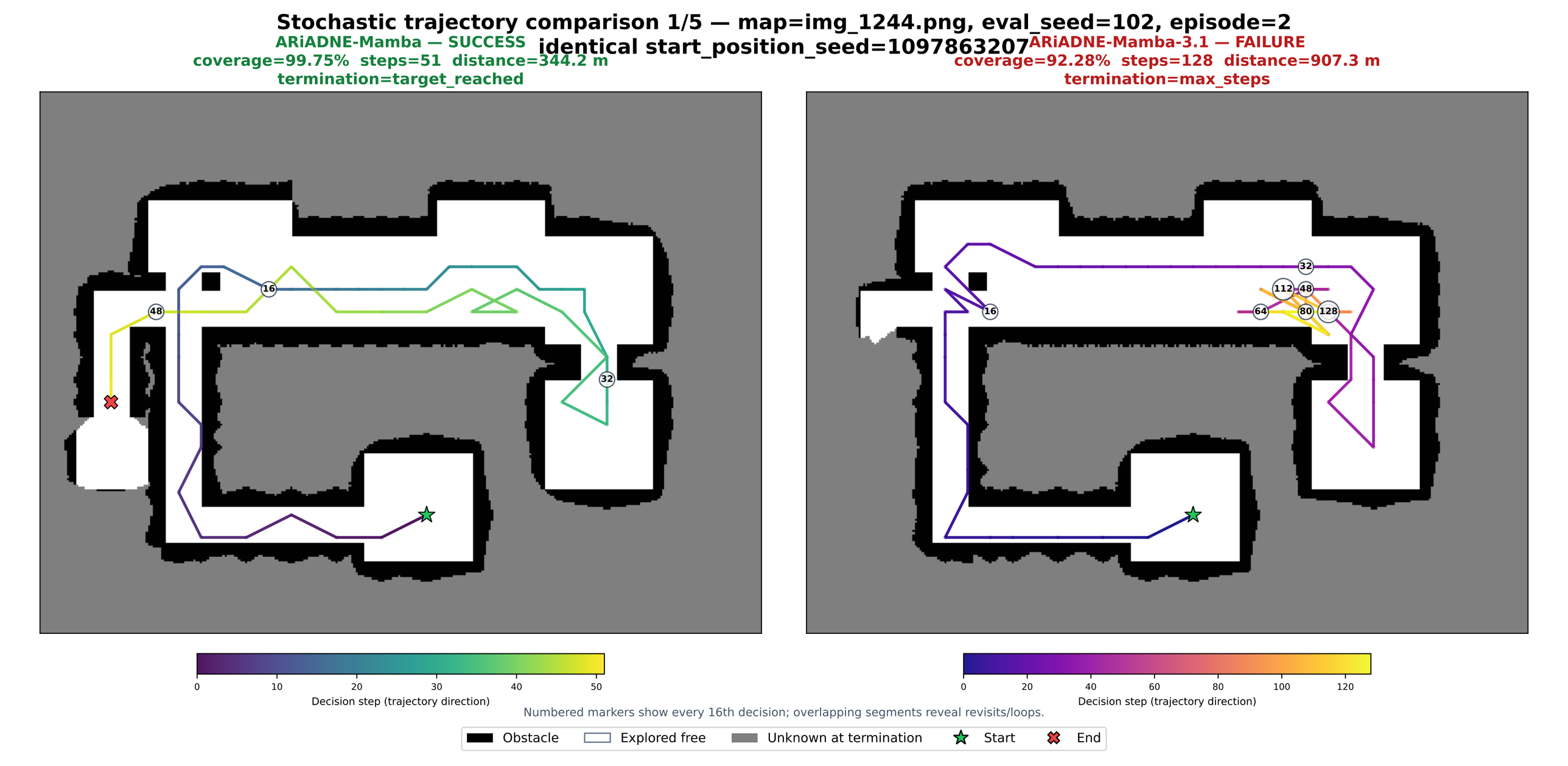
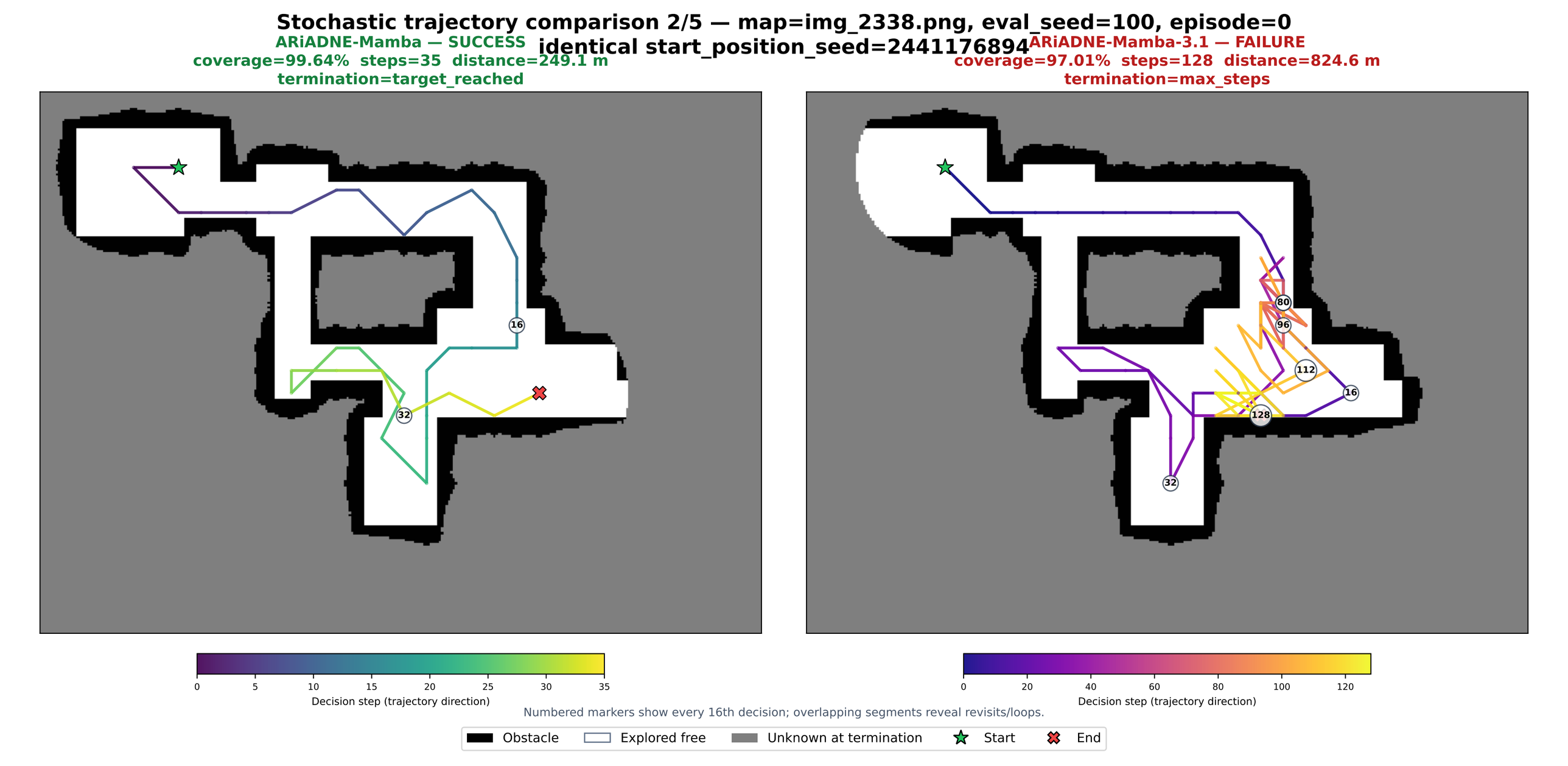
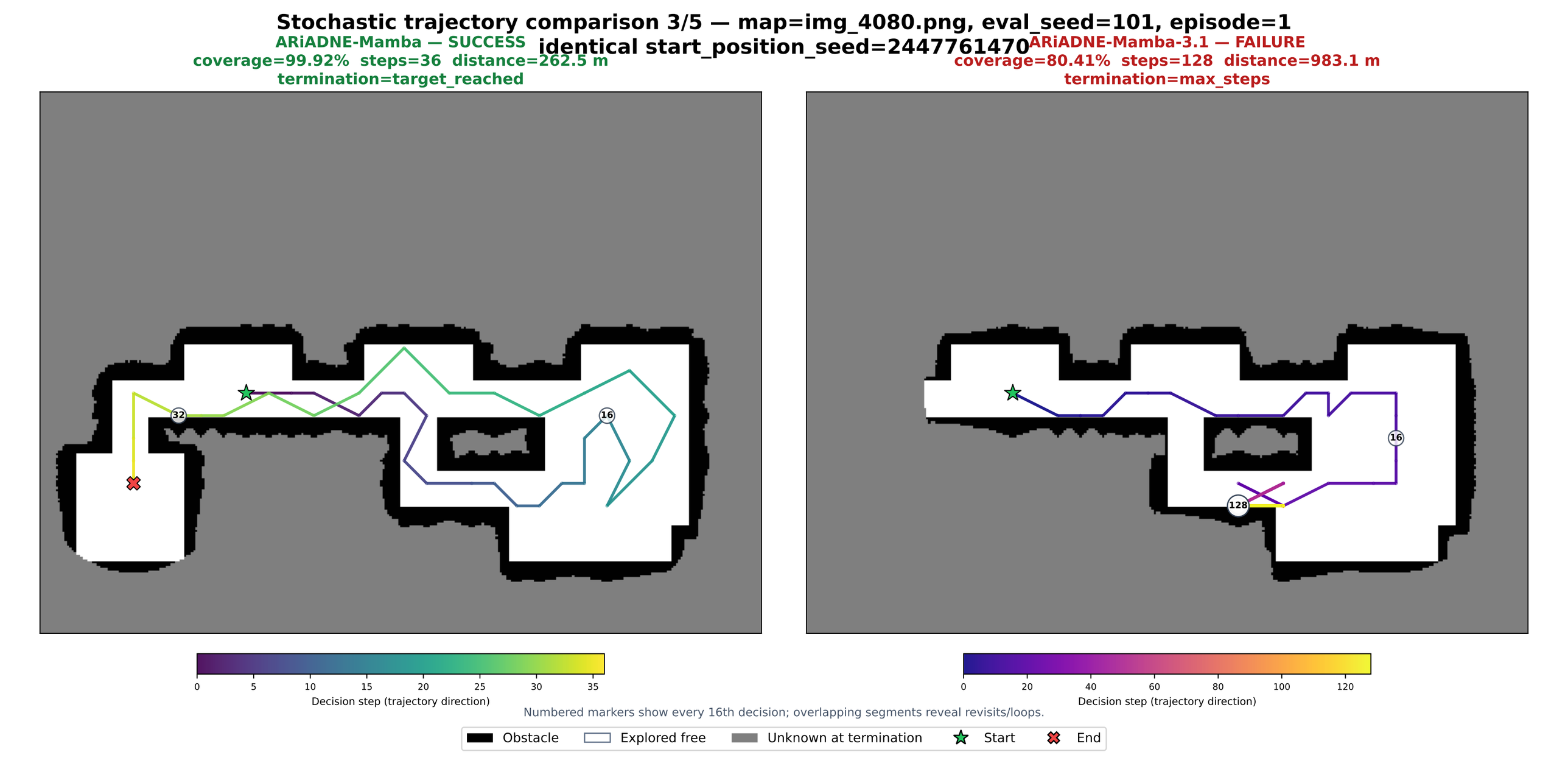
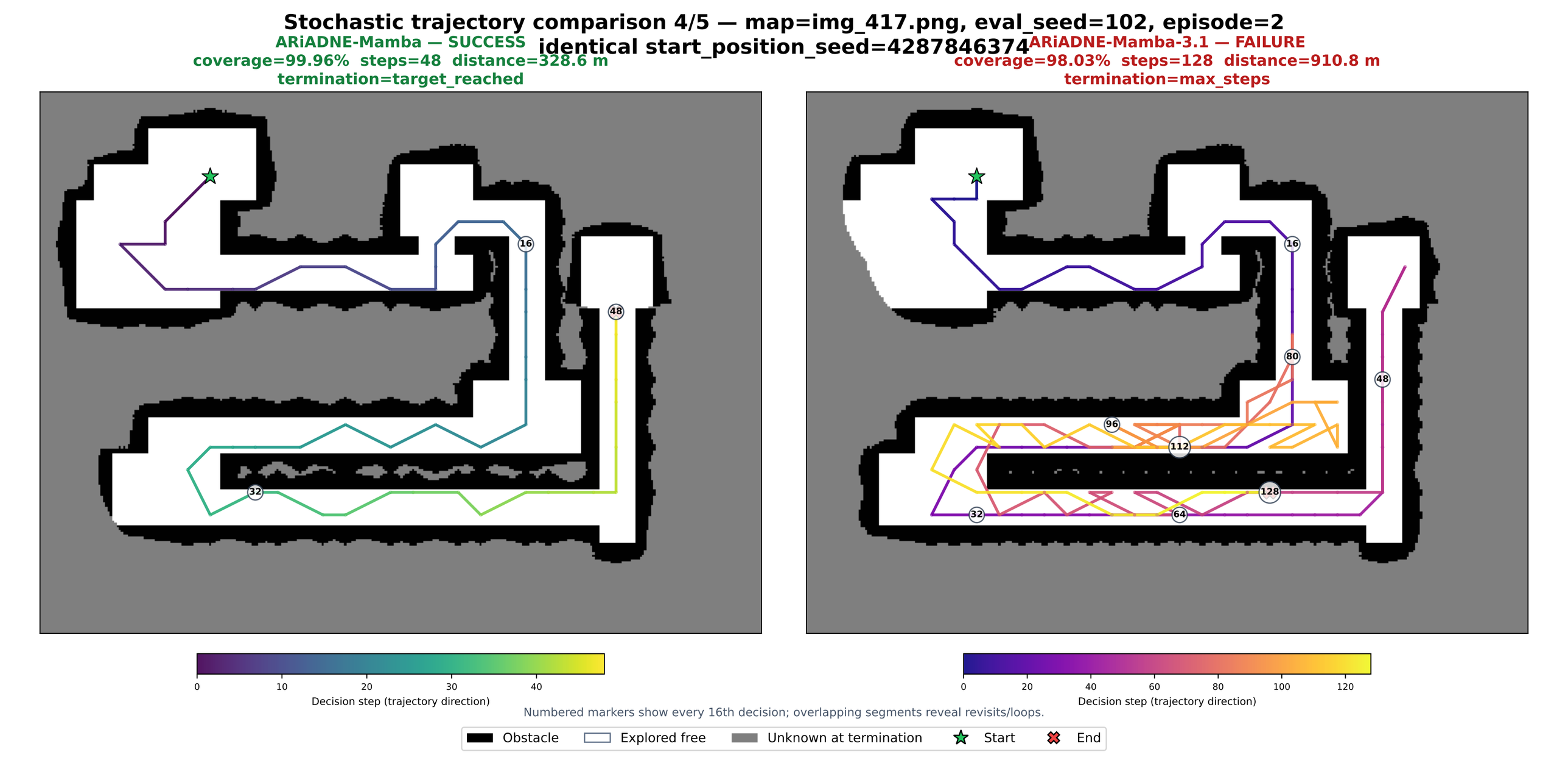

Stochastic trajectory comparison 1/5 — map=img_1244.png, eval_seed=102, episode=2
identical start_position_seed=1097863207
ARiADNE-Mamba — SUCCESS
coverage=99.75% steps=51 distance=344.2 m
termination=target_reached
ARiADNE-Mamba-3.1 — FAILURE
coverage=92.28% steps=128 distance=907.3 m
termination=max_steps
16
48
32
16
32
112
48
64
80
128
0
10
20
30
40
50
Decision step (trajectory direction)
0
20
40
60
80
100
120
Decision step (trajectory direction)
Numbered markers show every 16th decision; overlapping segments reveal revisits/loops.
Obstacle
Explored free
Unknown at termination
Start
End


Stochastic trajectory comparison 2/5 — map=img_2338.png, eval_seed=100, episode=0
identical start_position_seed=2441176894
ARiADNE-Mamba — SUCCESS
coverage=99.64% steps=35 distance=249.1 m
termination=target_reached
ARiADNE-Mamba-3.1 — FAILURE
coverage=97.01% steps=128 distance=824.6 m
termination=max_steps
16
32
80
96
112
16
128
32
0 5 10 15 20 25 30 35
Decision step (trajectory direction)
0 20 40 60 80 100 120
Decision step (trajectory direction)
Numbered markers show every 16th decision; overlapping segments reveal revisits/loops.
Obstacle
Explored free
Unknown at termination
Start
End


Stochastic trajectory comparison 3/5 — map=img_4080.png, eval_seed=101, episode=1
identical start_position_seed=2447761470
ARiADNE-Mamba — SUCCESS
coverage=99.92% steps=36 distance=262.5 m
termination=target_reached
ARiADNE-Mamba-3.1 — FAILURE
coverage=80.41% steps=128 distance=983.1 m
termination=max_steps
32
16
16
128
0 5 10 15 20 25 30 35
Decision step (trajectory direction)
0 20 40 60 80 100 120
Decision step (trajectory direction)
Numbered markers show every 16th decision; overlapping segments reveal revisits/loops.
Obstacle
Explored free
Unknown at termination
Start
End

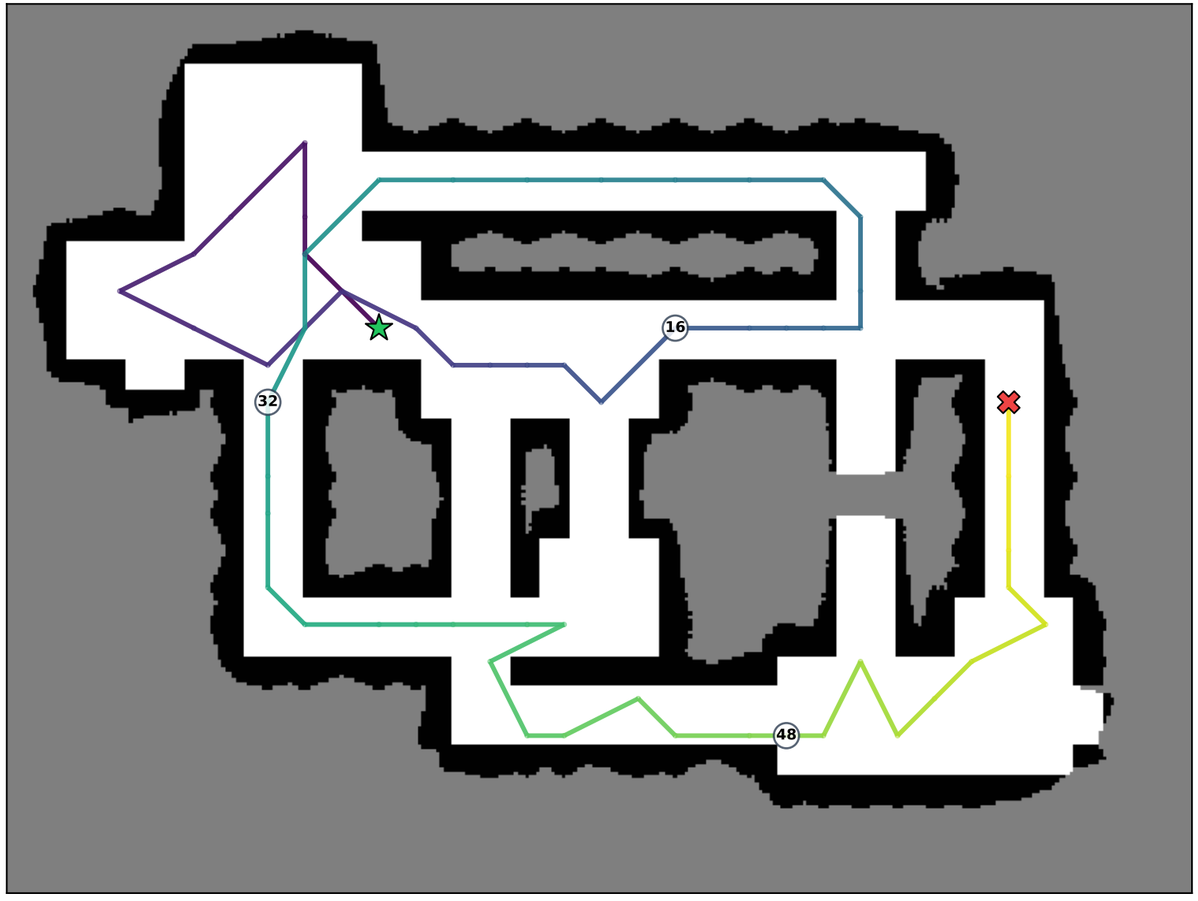

Stochastic trajectory comparison 4/5 — map=img_417.png, eval_seed=102, episode=2
identical start_position_seed=4287846374
ARiADNE-Mamba — SUCCESS
coverage=99.96% steps=48 distance=328.6 m
termination=target_reached
ARiADNE-Mamba-3.1 — FAILURE
coverage=98.03% steps=128 distance=910.8 m
termination=max_steps
16
48
32
16
80
48
96
112
128
32
64
0
10
20
30
40
Decision step (trajectory direction)
0
20
40
60
80
100
120
Decision step (trajectory direction)
Numbered markers show every 16th decision; overlapping segments reveal revisits/loops.
Obstacle
Explored free
Unknown at termination
Start
End

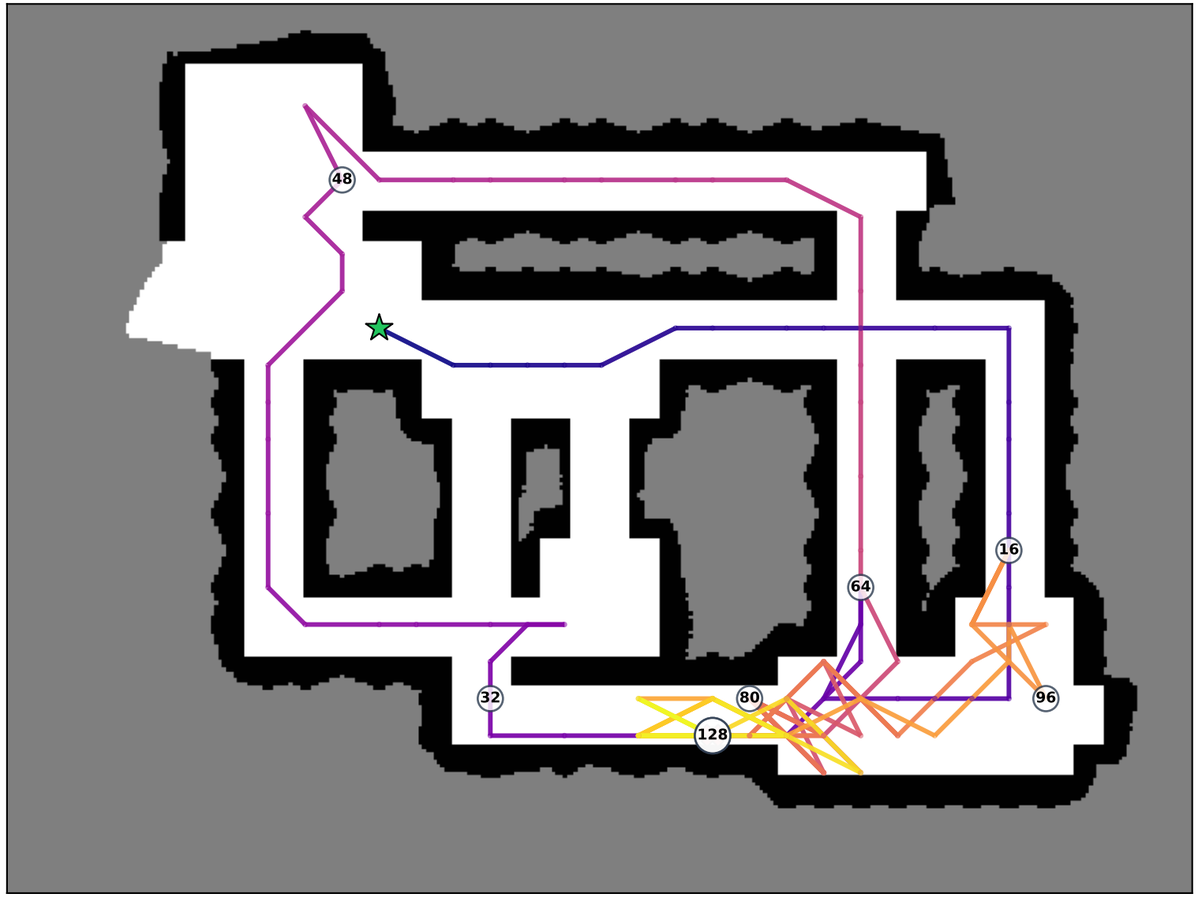
Stochastic trajectory comparison 5/5 — map=img_853.png, eval_seed=101, episode=1

identical start_position_seed=2380604721

ARiADNE-Mamba — SUCCESS
coverage=99.18% steps=58 distance=405.9 m
termination=target_reached

ARiADNE-Mamba-3.1 — FAILURE
coverage=96.35% steps=128 distance=1007.6 m
termination=max_steps





0 10 20 30 40 50
Decision step (trajectory direction)

0 20 40 60 80 100 120
Decision step (trajectory direction)

Numbered markers show every 16th decision; overlapping segments reveal revisits/loops.

Obstacle Explored free Unknown at termination Start End